Snowboard Rev 3 — RP2350A flight controller

Block diagram: each box is a sub-sheet (double-click to open). GND, +3V3 and +5V are global power nets and are not drawn here.

Power

BAT
BAT_F
VBUS

File: sheets/power.kicad_sch

BAT 9–25 V from ESC connector

TPS54202 buck -> 5 V 2 A

AP2112K LDO -> 3.3 V

USB VBUS OR-ed onto 5 V

USB

VBUS
USB_DM
USB_DP

File: sheets/usb.kicad_sch

USB-C 2.0 receptacle (J2)

USBLC6 ESD, 27 R series

5.1 k CC pull-downs

MCU

IMU_SCK
IMU_MOSI
IMU_MISO
IMU_CS
IMU_INT1
M1
M2
M3
M4
ESC_TEL
VBAT_SENSE
CUR_SENSE
RX_TX
RX_RX
AUX_TX
AUX_RX
I2C_SDA
I2C_SCL
BUZZ
RSSI
USB_DM
USB_DP
STATUS_LED
WS_DATA

File: sheets/mcu.kicad_sch

RP2350A + 4 MB QSPI flash

12 MHz crystal, 1.1 V core SMPS

BOOTSEL, RUN, SWD (J4)

IMU

IMU_SCK
IMU_MOSI
IMU_MISO
IMU_CS
IMU_INT1

File: sheets/imu.kicad_sch

ICM-42688-P, SPI0 @ 24 MHz

INT1 interrupt to GPIO6

ferrite-filtered 3.3 V

IO

M1
M2
M3
M4
ESC_TEL
VBAT_SENSE
CUR_SENSE
RX_TX
RX_RX
AUX_TX
AUX_RX
I2C_SDA
I2C_SCL
BUZZ
RSSI
BAT
BAT_F

File: sheets/io.kicad_sch

J1 ESC 8-pin SH (SpeedyBee order)

J3 RX 4-pin SH (EP2, CRSF)

VBAT / current sense, buzzer, pads

LEDs

STATUS_LED
WS_DATA

File: sheets/leds.kicad_sch

Status LED

XL-1615 RGB LED via 74AHCT1G125

LED strip output pads

ESC J1 = SpeedyBee F405 V4 FC-side order; RX J3 = EP2 (5V GND TX RX)
Spec: SNB-HW-001. RP2350 core per Raspberry Pi RP2350A Minimal R4.

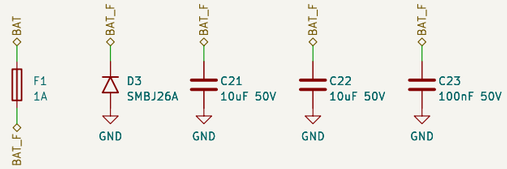
Snowboard Rev 3 — RP2350A flight controller (snowboard_fc)

Sheet: /
File: snowboard_fc.kicad_sch

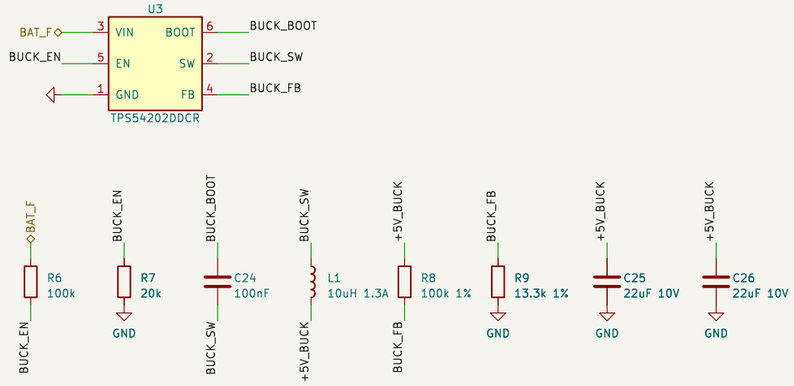
Title: Snowboard flight controller — top level

Size: A3	Date: 2026-10-16	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/7

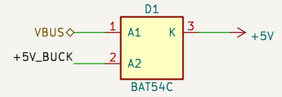
Battery input protection



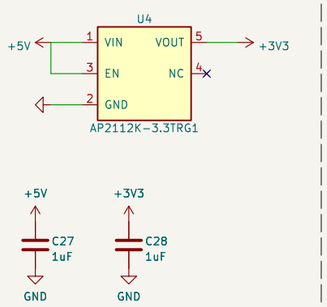
5 V buck — TPS54202



USB / buck OR-ing



3.3 V LDO — AP2112K



Power flags (ERC drivers)



Spec: SNB-HW-001. RP2350 core per Raspberry Pi RP2350A Minimal R4.

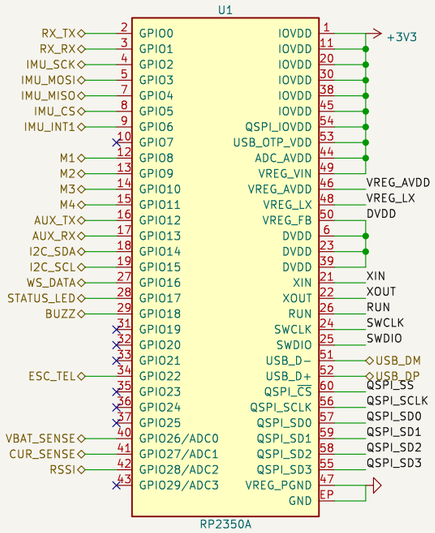
Snowboard Rev 3 — RP2350A flight controller (snowboard_fc)

Sheet: /Power/
File: power.kicad_sch

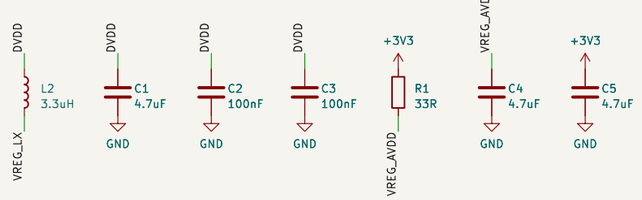
Title: Power — input protection, 5 V buck, 3.3 V LDO

Size: A3	Date: 2026-10-16	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/7

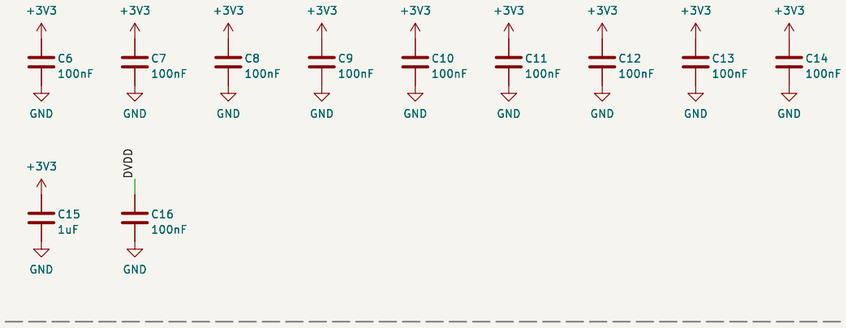
RP2350A



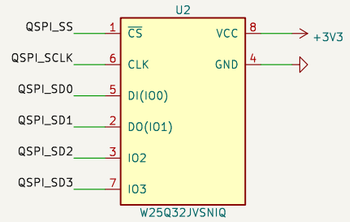
1.1 V core regulator (on-chip SMPS)



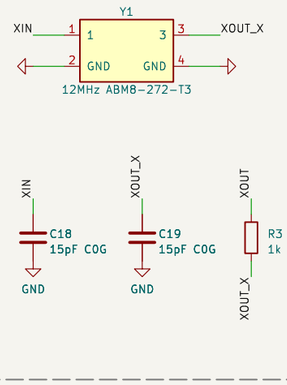
3.3 V decoupling — 100 nF at each IOVDD / QSPI / OTP / ADC pin



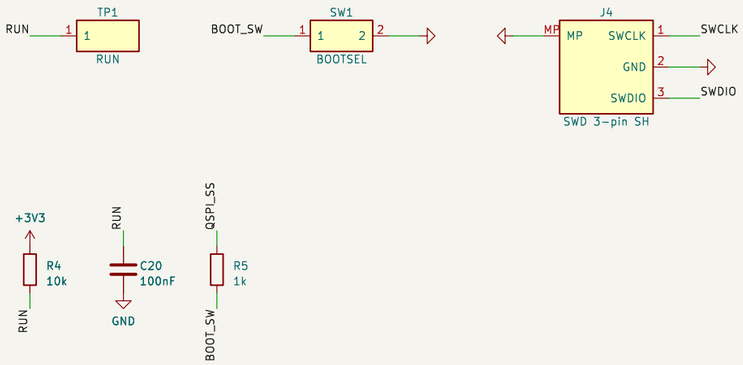
QSPI flash



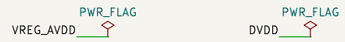
12 MHz crystal



Reset, BOOTSEL & SWD



Power flags (ERC drivers)



Spec: SNB-HW-001. RP2350 core per Raspberry Pi RP2350A Minimal R4.

Snowboard Rev 3 — RP2350A flight controller (snowboard_fc)

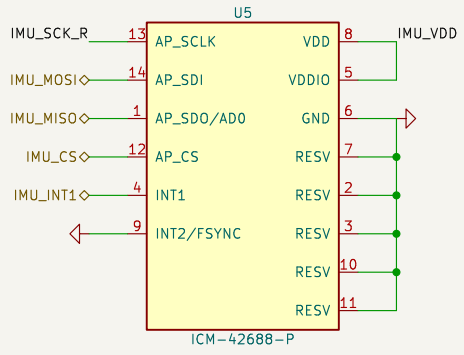
Sheet: /MCU/

File: mcu.kicad_sch

Title: MCU — RP2350A, flash, clock, debug

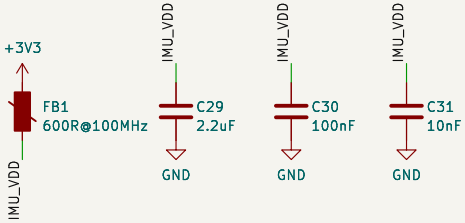
Size: A3	Date: 2026-10-16	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7

ICM-42688-P





Filtered supply (DS-000347 Table 11)



Power flags (ERC drivers)



Spec: SNB-HW-001. RP2350 core per Raspberry Pi RP2350A Minimal R4.

Snowboard Rev 3 — RP2350A flight controller (snowboard_fc)

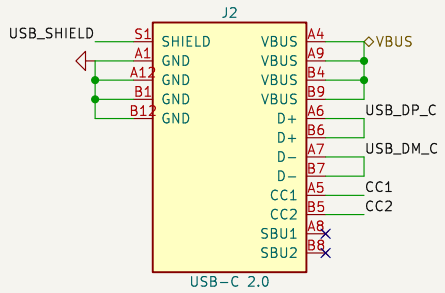
Sheet: /IMU/

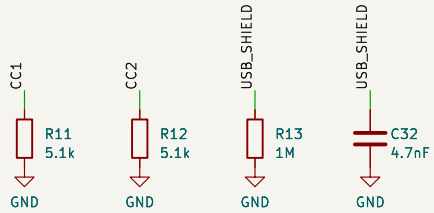
File: imu.kicad_sch

Title: IMU — ICM-42688-P on SPI0

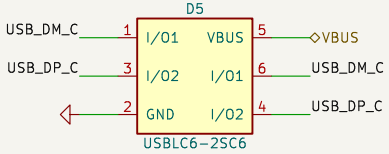
Size: A4	Date: 2026-10-16	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

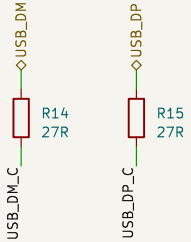
USB-C receptacle





ESD protection & series termination





Spec: SNB-HW-001. RP2350 core per Raspberry Pi RP2350A Minimal R4.

Snowboard Rev 3 — RP2350A flight controller (snowboard_fc)

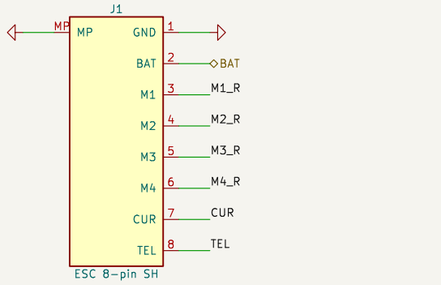
Sheet: /USB/

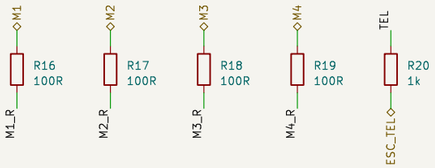
File: usb.kicad_sch

Title: USB-C — flashing and debug console

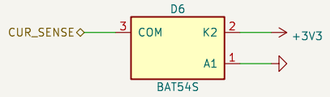
Size: A4	Date: 2026-10-16	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/7

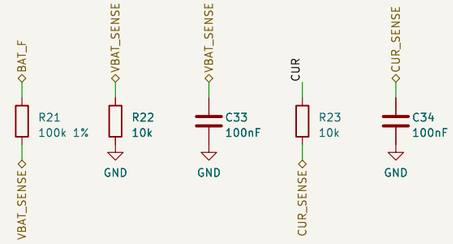
ESC connector — SpeedyBee 8-pin



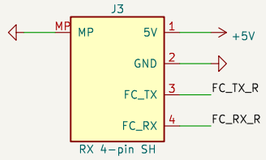


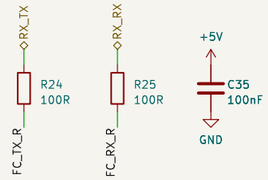
Battery voltage & current sense



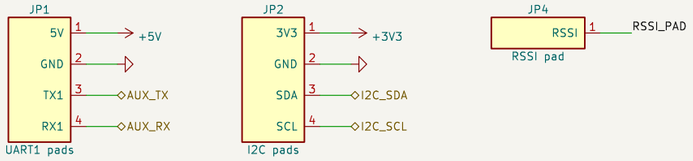


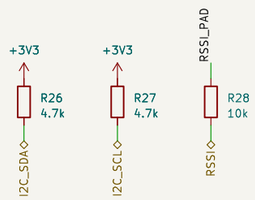
RC receiver — EP2 ELRS (CRSF)



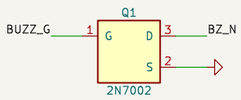


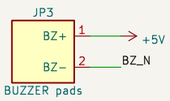
Aux UART, I2C & RSSI pads

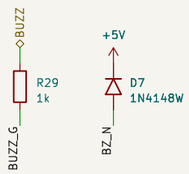




Buzzer driver







Spec: SNB-HW-001. RP2350 core per Raspberry Pi RP2350A Minimal R4.

Snowboard Rev 3 — RP2350A flight controller (snowboard_fc)

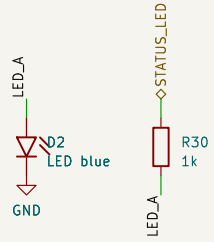
Sheet: /IO/

File: io.kicad_sch

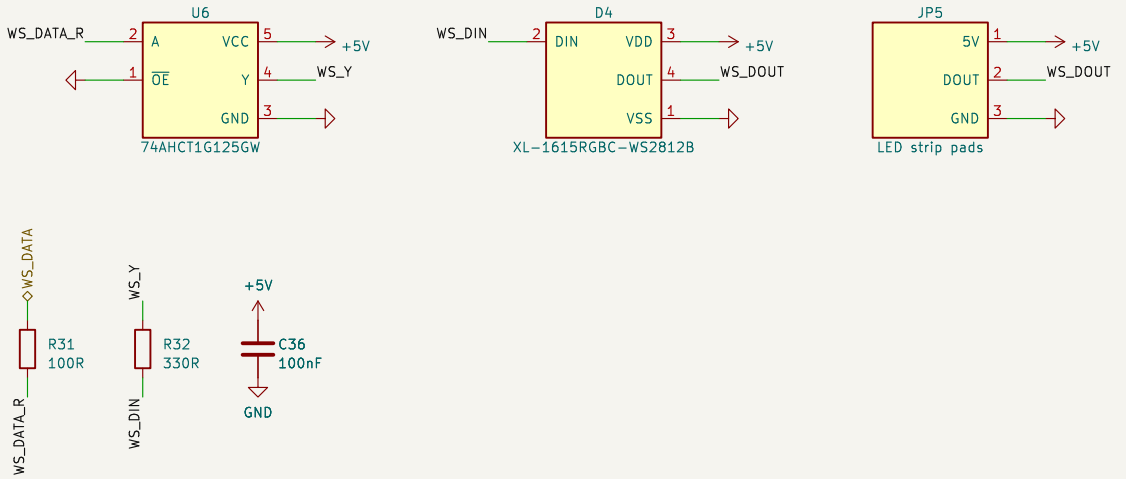
Title: ESC, RC receiver and auxiliary I/O

Size: A3	Date: 2026-10-16	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/7

Status LED



XL-1615RGBC-WS2812B + 5 V level shifter



Spec: SNB-HW-001. RP2350 core per Raspberry Pi RP2350A Minimal R4.

Snowboard Rev 3 — RP2350A flight controller (snowboard_fc)

Sheet: /LEDs/
File: leds.kicad_sch

Title: LEDs — status LED and XL-1615 RGB LED

Size: A4	Date: 2026-10-16	**Rev: 3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/7